Responsible AI Framework: Explainability & Transparency Design Considerations

NovaCare Health Systems

Course: Responsible Appl of AI (AIML-510-01R4A)

Assignment: 2.4 RAI Framework Project Design Considerations

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NovaCare Health Systems

Fictional healthcare enterprise (consistent with Assignments 1.3 and 1.4)

- **Scale** — 30 hospitals, 150+ outpatient facilities, approximately 45,000 employees
- **Reach** — Serves about 6 million patients annually through hospitals, clinics, diagnostic centers, and telehealth
- **Technology footprint** — Enterprise EHR, cloud analytics, imaging, patient portals, and digital care platforms
- **RAI foundation** — Seven ethical principles established in 1.3, operationalized through the NIST AI RMF (Govern, Map, Measure, Manage) in 1.4

AI Use Cases Under Consideration

Same four use cases established in Assignment 1.3

1

AI-Assisted Medical Imaging

Detects abnormalities in X-rays, CT scans, and MRIs

2

Patient Deterioration Prediction

Predicts patients at risk of rapid clinical decline

3

Generative AI Clinical Assistant

Summarizes records and drafts clinical documentation

4

AI Scheduling & Resource Optimization

Forecasts demand and optimizes staffing and appointments

Chart: Explainability Need by Use Case

Which use cases require strong explainability, and why

Use Case	Explainability Need	Rationale
AI-Assisted Medical Imaging	**High**	Directly informs diagnosis. Radiologists need local explanations to catch errors like atypical findings; governance needs global explanations to validate the tool before deployment.
Patient Deterioration Prediction	**High**	Triggers clinical intervention. A missed or false flag affects patient safety, so clinicians need to know why a patient was flagged to decide how urgently to act.
Generative AI Clinical Assistant	**Moderate**	Drafts are reviewed by a clinician before use, but hallucination risk means clinicians need enough visibility into sourcing to catch errors before signing off.
AI Scheduling & Resource Optimization	**Low**	Does not affect individual patient treatment. Accuracy matters more than explainability, though enough transparency should exist to explain staffing decisions if disputed.

Chart: Process for Evaluating AI Transparency/Explainability

Six-step evaluation path, tied to the RAI roles defined in Assignment 1.4

Step	Activity	Owner
1. Risk Tiering	Set explainability requirement based on the risk tier from the prior chart	*Use-Case Owner, AI Risk & Compliance Manager*
2. Method Selection	Choose global vs. local explanation methods matched to the audience	*AI Safety & Validation Lead, Human Factors & Clinical AI Lead*
3. Usability Testing	Test explanations with real end users to confirm they are true, useful, and understandable	*Human Factors & Clinical AI Lead*
4. Independent Validation Gate	Independent review confirms explainability meets the validation gate before approval	*AI Safety & Validation Lead*
5. RAI Approval	Steering Committee sign-off required for high-risk systems	*Chief Responsible AI Officer, RAI Steering Committee*
6. Post-Deployment Monitoring	Track clinician override rates and complaints as an ongoing signal that explanations are working	*RAI Governance & Ethics Lead*

Chart: Documentation Across Development Steps

What transparency artifact is produced at each stage of the 1.4 vetting process

Development Stage	Documentation Produced	Purpose
Intake & Use-Case Owner	Use-case charter (intended purpose, affected populations)	*Defines scope for future explanations*
Risk Tier & Impact Map	Risk tier and explainability requirement	*Sets the level of explanation the system will need*
Data & Privacy Assessment	Data lineage record	*Documents inputs; supports global explanations*
Build + Documentation	Model card/fact sheet: capabilities, limitations, training data, subgroup performance	*Central transparency artifact (Google Model Card Toolkit, IBM AI FactSheets)*
Independent Validation	Validation report covering explainability and usability testing	*Confirms explanations work for their intended audience*
RAI Approval	Approval record with named executive owner	*Ties transparency to Accountability*
Deploy + Monitor	Clinician-facing transparency notice plus ongoing monitoring log	*Keeps end users informed as the system operates*

Conclusion

Difficulties and open questions from this workshop

- Chapter 3's biggest challenge for me: there's no formula for how much accuracy you should give up for explainability. The book just says "it depends," so applying that to a real use case like patient deterioration prediction is tougher than it sounds.
- Figuring out how much explanation is actually "enough" is harder than I expected. Too little and clinicians can't use it, too much and it slows them down or gives false confidence. There's no clear line for where that sits.
- Turning a principle like Fairness & Equity into an actual measurable KPI feels like a big jump. It's easy to say we care about it, harder to say who owns turning that into a number.
- Open question: who should actually decide how much explanation is "enough" for a use case? The clinicians using it, the ethics committee, or the vendor?
- Open question: if the Generative AI Clinical Assistant is always reviewed by a human anyway, does it really need the same explainability standard as imaging or deterioration prediction, or is that human check enough on its own?

References

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- National Institute of Standards and Technology. (2023). Artificial intelligence risk management framework (AI RMF 1.0) (NIST AI 100-1). U.S. Department of Commerce.